GraphTheorySymbol Memo
v1.4.0 2026/09/12

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
`\GTSadjacency/`	`\GTSa/`		u v
`\GTSadjacencyRelation/`	`\GTSaR/`		$u\ v$
`\GTSunadjacency/`	`\GTSu/`		u v
`\GTSunadjacencyRelation/`	`\GTSuR/`		$u\ v$
`\GTSdirectedAdjacencyUp/`	`\GTSdAU/`		u v
`\GTSdirectedAdjacencyUpRelation/`	`\GTSdAUR/`		$u\ v$
`\GTSnegatedDirectedAdjacencyUp/`	`\GTSnDAU/`		u v
`\GTSnegatedDirectedAdjacencyUpRelation/`	`\GTSnDAUR/`		$u\ v$
`\GTSdirectedAdjacencyDown/`	`\GTSdAD/`		u v
`\GTSdirectedAdjacencyDownRelation/`	`\GTSdADR/`		$u\ v$
`\GTSnegatedDirectedAdjacencyDown/`	`\GTSnDAD/`		u v
`\GTSnegatedDirectedAdjacencyDownRelation/`	`\GTSnDADR/`		$u\ v$
`\GTSincidence/`	`\GTSi/`		v e
`\GTSincidenceRelation/`	`\GTSiR/`		$v\ e$
`\GTSnegatedIncidence/`	`\GTSnI/`		v e
`\GTSnegatedIncidenceRelation/`	`\GTSnIR/`		$v\ e$
`\GTSoutIncidence/`	`\GTSoI/`		v e
`\GTSoutIncidenceRelation/`	`\GTSoIR/`		$v\ e$
`\GTSnegatedOutIncidence/`	`\GTSnOI/`		v e
`\GTSnegatedOutIncidenceRelation/`	`\GTSnOIR/`		$v\ e$
`\GTSinIncidence/`	`\GTSiI/`		v e
`\GTSinIncidenceRelation/`	`\GTSiIR/`		$v\ e$
`\GTSnegatedInIncidence/`	`\GTSnII/`		v e
`\GTSnegatedInIncidenceRelation/`	`\GTSnIIR/`		$v\ e$

Long name	Short name	Result	In context
`\GTSadjacencyMonochrome/`	`\GTSaM/`		u v
`\GTSadjacencyMonochromeRelation/`	`\GTSaMR/`		u v
`\GTSunadjacencyMonochrome/`	`\GTSuM/`		u v
`\GTSunadjacencyMonochromeRelation/`	`\GTSuMR/`		u v
`\GTSdirectedAdjacencyUpMonochrome/`	`\GTSdAUM/`		u v
`\GTSdirectedAdjacencyUpMonochromeRelation/`	`\GTSdAUMR/`		u v
`\GTSnegatedDirectedAdjacencyUpMonochrome/`	`\GTSnDAUM/`		u v
`\GTSnegatedDirectedAdjacencyUpMonochromeRelation/`	`\GTSnDAUMR/`		u v
`\GTSdirectedAdjacencyDownMonochrome/`	`\GTSdADM/`		u v
`\GTSdirectedAdjacencyDownMonochromeRelation/`	`\GTSdADMR/`		u v
`\GTSnegatedDirectedAdjacencyDownMonochrome/`	`\GTSnDADM/`		u v
`\GTSnegatedDirectedAdjacencyDownMonochromeRelation/`	`\GTSnDADMR/`		u v
`\GTSincidenceMonochrome/`	`\GTSiM/`		v e
`\GTSincidenceMonochromeRelation/`	`\GTSiMR/`		v e
`\GTSnegatedIncidenceMonochrome/`	`\GTSnIM/`		v e
`\GTSnegatedIncidenceMonochromeRelation/`	`\GTSnIMR/`		v e
`\GTSoutIncidenceMonochrome/`	`\GTSoIM/`		v e
`\GTSoutIncidenceMonochromeRelation/`	`\GTSoIMR/`		v e
`\GTSnegatedOutIncidenceMonochrome/`	`\GTSnOIM/`		v e
`\GTSnegatedOutIncidenceMonochromeRelation/`	`\GTSnOIMR/`		v e
`\GTSinIncidenceMonochrome/`	`\GTSiIM/`		v e
`\GTSinIncidenceMonochromeRelation/`	`\GTSiIMR/`		v e
`\GTSnegatedInIncidenceMonochrome/`	`\GTSnIIM/`		v e
`\GTSnegatedInIncidenceMonochromeRelation/`	`\GTSnIIMR/`		v e

```
\GTSsetMainColor{blue}
```

Long name	Short name	Result	In context
`\GTSadjacency/`	`\GTSa/`		u v
`\GTSadjacencyRelation/`	`\GTSaR/`		u v
`\GTSunadjacency/`	`\GTSu/`		u v
`\GTSunadjacencyRelation/`	`\GTSuR/`		u v
`\GTSdirectedAdjacencyUp/`	`\GTSdAU/`		u v
`\GTSdirectedAdjacencyUpRelation/`	`\GTSdAUR/`		u v
`\GTSnegatedDirectedAdjacencyUp/`	`\GTSnDAU/`		u v
`\GTSnegatedDirectedAdjacencyUpRelation/`	`\GTSnDAUR/`		u v
`\GTSdirectedAdjacencyDown/`	`\GTSdAD/`		u v
`\GTSdirectedAdjacencyDownRelation/`	`\GTSdADR/`		u v
`\GTSnegatedDirectedAdjacencyDown/`	`\GTSnDAD/`		u v
`\GTSnegatedDirectedAdjacencyDownRelation/`	`\GTSnDADR/`		u v
`\GTSincidence/`	`\GTSi/`		v e
`\GTSincidenceRelation/`	`\GTSiR/`		v e
`\GTSnegatedIncidence/`	`\GTSnI/`		v e
`\GTSnegatedIncidenceRelation/`	`\GTSnIR/`		v e
`\GTSoutIncidence/`	`\GTSoI/`		v e
`\GTSoutIncidenceRelation/`	`\GTSoIR/`		v e
`\GTSnegatedOutIncidence/`	`\GTSnOI/`		v e
`\GTSnegatedOutIncidenceRelation/`	`\GTSnOIR/`		v e
`\GTSinIncidence/`	`\GTSiI/`		v e
`\GTSinIncidenceRelation/`	`\GTSiIR/`		v e
`\GTSnegatedInIncidence/`	`\GTSnII/`		v e
`\GTSnegatedInIncidenceRelation/`	`\GTSnIIR/`		v e

Long name	Short name	Result	In context
`\GTSmetaNot/`	`\GTSmN/`	$\overset{\mathrm{m}}{\neg}$	$\overset{\mathrm{m}}{\neg}$
`\GTSmetaAnd/`	`\GTSmA/`	$\overset{\mathrm{m}}{\wedge}$	$\overset{\mathrm{m}}{\wedge}$
`\GTSmetaOr/`	`\GTSmO/`	$\overset{\mathrm{m}}{\vee}$	$\overset{\mathrm{m}}{\vee}$

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:
Tournament adjacency:
$\forall u, v, u\ [\overset{\mathrm{m}}{\vee}]\ v.$
Tournament adjacency with maybe some colored edge also, who knows?:
$\forall u, v, u\ \overset{\mathrm{m}}{\vee}\ v.$
Directed graph:
$\forall u, v, u\ [\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\vee}]\ v.$
Oriented graph:
$\forall u, v, u\ [\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\vee}]\ v.$

Prefix notation:
$\exists u, v, u\ [\overset{\mathrm{m}}{\vee}]\ v.$
$\exists v, e, v\ [\overset{\mathrm{m}}{\vee}]\ e.$